

Major Project Presentation (Phase I)

Project Review 2 On

An Explainable Deep Learning Framework For Diabetic Retinopathy Severity Grading

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Team Work Division

Member	Area	Main Responsibilities
R. Ram Charan (23R21A66H6)	Research & Dataset collection	Literature survey, research gap, clinical dataset study and dataset organization
K. Angel (23R21A66F7)	Image data Processing	Fundus image collection, image cleaning, resizing, noise removal, and normalization
J. Pravalika (23R21A66F3)	Deep Learning Model	EfficientNet-B0, transfer learning, feature extraction, model training and DR classification
G. Sainath (23R21A66F0)	Explainable AI	Grad-CAM, heatmap generation, lesion visualization and prediction explanation
K. Karthik (23R21A66K1)	Evaluation & System Integration	Accuracy, Precision, Recall, F1-Score, QWK, confidence scores and result analysis

Introduction

Diabetic Retinopathy (DR) is one of the leading causes of preventable blindness among people with diabetes. Early detection is essential to prevent vision loss, but manual screening of retinal fundus images is time-consuming and depends on the availability of experienced ophthalmologists, especially in underserved regions. Although deep learning models have demonstrated high accuracy in automated DR severity grading, their lack of interpretability limits clinical trust and adoption. This project proposes an Explainable Deep Learning Framework for Diabetic Retinopathy Severity Grading that automatically classifies DR severity while employing Explainable AI techniques, such as Grad-CAM, to highlight retinal lesions including microaneurysms, hemorrhages, and exudates that influence the model's predictions. This enhances transparency, assists clinicians in validating AI decisions, and supports reliable, scalable, and early diabetic retinopathy screening.

Abstract

Diabetic Retinopathy (DR) is one of the leading causes of preventable blindness among people with diabetes. Early detection is crucial, but manual screening of retinal fundus images is time-consuming and depends on expert ophthalmologists. This project presents an Explainable Deep Learning Framework for Automated Diabetic Retinopathy Severity Grading using EfficientNet-B0 and Grad-CAM. The proposed model classifies retinal images into five DR severity levels and highlights the retinal regions influencing its predictions, improving transparency and clinical trust. The framework is trained on the retinal funds image dataset and evaluated using Accuracy, Precision, Recall, F1-Score, and Quadratic Weighted Kappa(QWK). The proposed system supports accurate, efficient, and explainable diabetic retinopathy screening.

Keywords: Diabetic Retinopathy, Deep Learning, Explainable AI (XAI), EfficientNet-B0, Grad-CAM, Retinal Fundus Images, Severity Grading

Literature Survey

No	Title	Author	Journal Name & Year	Methodology Adapted	Key Findings	Gaps
1	A Clinically Interpretable Deep Learning Pipeline for Diabetic Retinopathy Classification Using EfficientNet and GradCAM	K. K. Ghadai, S. K. Nayak, B. R. Senapati	BMC Ophthalmology, 2026	EfficientNet-based feature extraction, combined with data augmentation and Grad-CAM, will be incorporated into our framework for DR severity grading.	Combines DR classification with visual explanations of important retinal regions.	Class imbalance and differences between adjacent DR stages remain challenging.
2	A Clinically Interpretable Deep Learning Pipeline for Diabetic Retinopathy Classification Using EfficientNet, Advanced Data Augmentation and GradCAM	Rashid Ali, Fiaz Gul Khan, Zia Ur Rehman, Daehan Kwak, Farman Ali	IEEE Journal of Biomedical and Health Informatics, 2025	The project takes the Explainable AI approach from this work to make model predictions more understandable and transparent.	Uses unlabeled data and Grad-CAM to improve DR detection and interpretability.	Unreliable pseudo-labels and imbalanced data can affect model learning

Literature Survey

No	Title	Author	Journal Name & Year	Methodology Adapted	Key Findings	Gaps
3	Diabetic Retinopathy Assessment through Multitask Learning Approach on Heterogeneous Fundus Image Datasets	Hongkang Wu, Kai Jin, Yiyang Jing, Wenyue Shen, Yih Chung Tham, Xiangji Pan	Ophthalmology Science, 2025	Clinical fundus image analysis and multitask learning concepts are considered to support training with real-world retinal images.	Uses clinical fundus images with ophthalmologist annotations and external validation.	Small sample size for some lesion types and limited representation across countries/ethnic groups.
4	Smart Grading of Diabetic Retinopathy: An Intelligent Recommendation-Based Fine-Tuned EfficientNetB0 Framework	Vatsala Anand, Deepika Koundal, Wael Y. Alghamdi, Bayan M. Alsharbi	Frontiers in Artificial Intelligence, 2024	Transfer learning using EfficientNet-B0 forms the basis of our deep learning model for five-level DR classification.	Shows the effectiveness of EfficientNet-B0 for automated DR grading.	Performance can vary across datasets; large-scale validation and fine-tuning are required.

Objectives

- To develop an automated deep learning model using EfficientNet-B0 with transfer learning for classifying retinal fundus images into five diabetic retinopathy severity levels.
- To improve the interpretability of the model by integrating Grad-CAM to visualize retinal regions influencing the classification.
- To develop an accurate and explainable clinical decision-support system for early diabetic retinopathy screening and severity grading.

Existing Systems

1.CNN-Based Diabetic Retinopathy Detection : A widely used deep learning approach that analyzes retinal fundus images using CNNs to detect diabetic retinopathy. It automatically learns important retinal features, but the predictions are often difficult to interpret.

2.Transfer Learning-Based DR Grading : A deep learning approach using pre-trained models such as ResNet, DenseNet, EfficientNet, and Inception to classify diabetic retinopathy into different severity levels. However, these systems may struggle with distinguishing between similar severity stages.

3.Explainable AI-Based DR Classification : An approach that combines deep learning with techniques such as Grad-CAM and Grad-CAM++ to highlight regions responsible for predictions. However, the explanations may be coarse and may not always provide clinically meaningful reasoning.

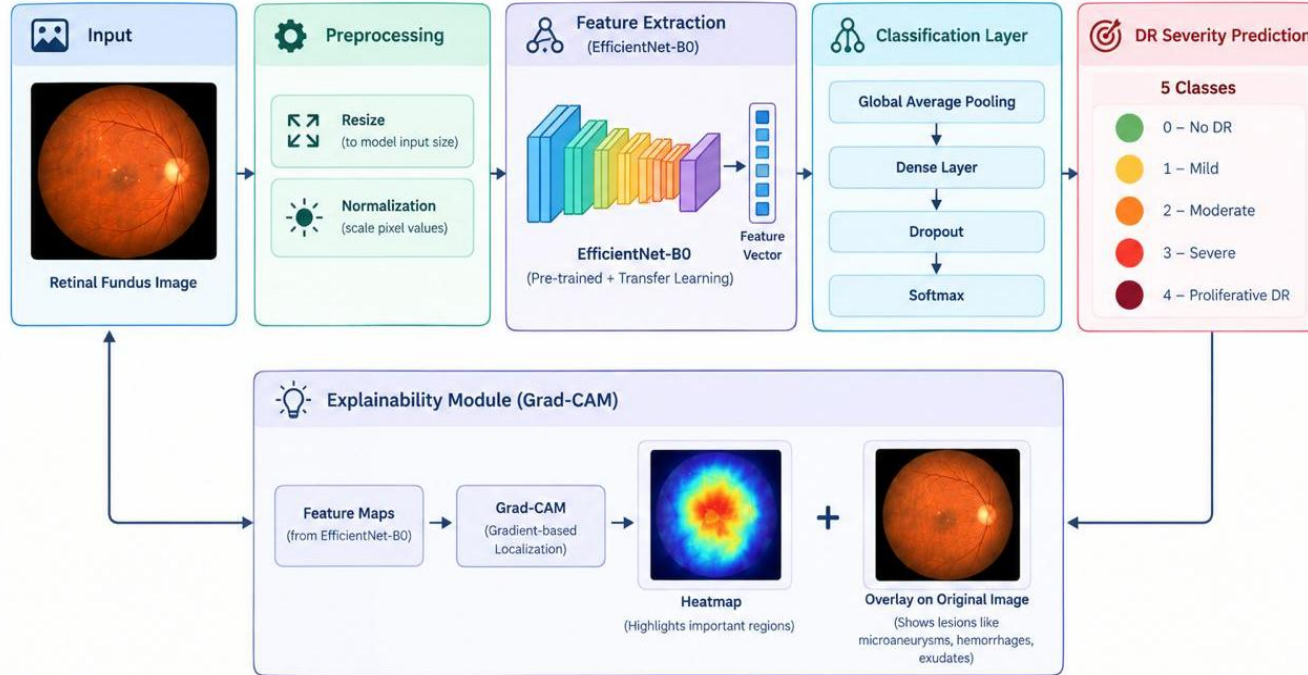
Disadvantages of Existing Systems

- **Limited Interpretability of Deep Learning Models:** Most existing diabetic retinopathy grading systems use complex deep learning models that provide accurate predictions but do not clearly explain the reasons behind their decisions. This makes it difficult for ophthalmologists to understand and trust the model's predictions.
- **Difficulty in Distinguishing DR Severity Levels:** Existing systems may struggle to differentiate between adjacent stages such as Mild, Moderate, and Severe diabetic retinopathy because the visual differences between these stages can be subtle. This can lead to incorrect severity grading.
- **Dependence on Dataset Quality and Class Imbalance:** Existing models are highly dependent on the quality and diversity of retinal fundus images used for training. Imbalanced datasets, poor-quality images, and variations in lighting or imaging devices can reduce the reliability and generalization of the predictions.

Proposed Methodology

The proposed system uses an explainable deep learning framework to automatically grade diabetic retinopathy severity from retinal fundus images. The input image is first preprocessed using resizing, noise removal, contrast enhancement, and normalization to improve image quality. The processed image is then provided to a transfer-learning-based deep learning model such as EfficientNet or ResNet, which extracts important retinal features and classifies the image into five severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR. To make the prediction interpretable, Explainable AI techniques such as Grad-CAM or Grad-CAM++ are applied to generate heatmaps highlighting the retinal regions that influenced the model's decision. The system finally combines the predicted severity, confidence score, and visual explanation into an understandable report, helping medical professionals interpret and verify the automated prediction.

Proposed Architecture



Proposed Architecture of An Explainable Deep Learning Framework for Diabetic Retinopathy Severity Grading

Advantages of Proposed Methodology

- **Provides accurate severity grading** by classifying retinal fundus images into five stages: No DR, Mild, Moderate, Severe, and Proliferative DR.
- **Provides explainable predictions** using Grad-CAM or Grad-CAM++ to highlight the retinal regions that contribute to the model's decision, improving transparency for medical professionals.
- **Improves reliability of diagnosis** by combining deep learning predictions with confidence scores, helping identify uncertain or difficult cases.
- **Reduces manual screening effort** by automatically analyzing fundus images and generating a structured severity assessment, supporting faster and more consistent diabetic retinopathy screening.

Datasets

1. **EyePACS Dataset:** Large collection of retinal fundus images with diabetic retinopathy severity labels.
2. **APTOS 2019 Dataset:** Retinal images categorized into five diabetic retinopathy severity levels.
3. **Messidor Dataset:** Annotated retinal fundus images used for diabetic retinopathy detection and grading.
4. **IDRiD Dataset:** Retinal images containing diabetic retinopathy grades and annotations for retinal lesions.

Hardware Requirement

1. **Processor:** Intel Core i5 / AMD Ryzen 5 or above
2. **RAM:** 8 GB minimum, 16 GB recommended
3. **GPU:** NVIDIA GPU with 4 GB+ VRAM recommended for deep learning training
4. **Operating System:** Windows / Linux / macOS

Software Requirements

Category	Technology
Programming Language	Python
Deep Learning Framework	TensorFlow, Keras, PyTorch
Image Processing	OpenCV, PIL
Deep Learning Models	EfficientNet, ResNet, DenseNet
Explainable AI	Grad-CAM, Grad-CAM++
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Backend / Deployment	Flask / Streamlit
Development Tools	Git, GitHub

Implementation

```
CLASS_NAMES = {
    0: 'No DR',
    1: 'Mild',
    2: 'Moderate',
    3: 'Severe',
    4: 'Proliferative DR'
}

df['diagnosis_name'] = df['diagnosis'].map(CLASS_NAMES)

plt.figure(figsize=(8, 5))
ax = sns.countplot(data=df, x='diagnosis_name',
                   order=[CLASS_NAMES[i] for i in range(5)],
                   hue='diagnosis_name', legend=False, palette='viridis')
plt.title('Class Distribution - APTOS 2019')
plt.xlabel('Severity Grade')
plt.ylabel('Number of Images')
for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width()/2, p.get_height()),
               ha='center', va='bottom')
plt.tight_layout()
plt.savefig(f'{PROJECT_DIR}/outputs/class_distribution.png', dpi=150)
plt.show()
```

```
# Visual sanity check: before vs after preprocessing
sample_ids = df['id_code'].sample(4, random_state=7).values

fig, axes = plt.subplots(2, 4, figsize=(16, 8))
for j, img_id in enumerate(sample_ids):
    raw_path = f'{RAW_DIR}/train_images/{img_id}.png'
    raw = cv2.cvtColor(cv2.imread(raw_path), cv2.COLOR_BGR2RGB)
    processed = preprocess_image(raw_path)

    axes[0, j].imshow(raw)
    axes[0, j].set_title('Original')
    axes[0, j].axis('off')

    axes[1, j].imshow(processed)
    axes[1, j].set_title('Preprocessed')
    axes[1, j].axis('off')

plt.tight_layout()
plt.savefig(f'{PROJECT_DIR}/outputs/preprocessing_comparison.png', dpi=150)
plt.show()
print('Compare rows: CLAHE should make blood vessels and lesions visibly sharper.')
```

Results

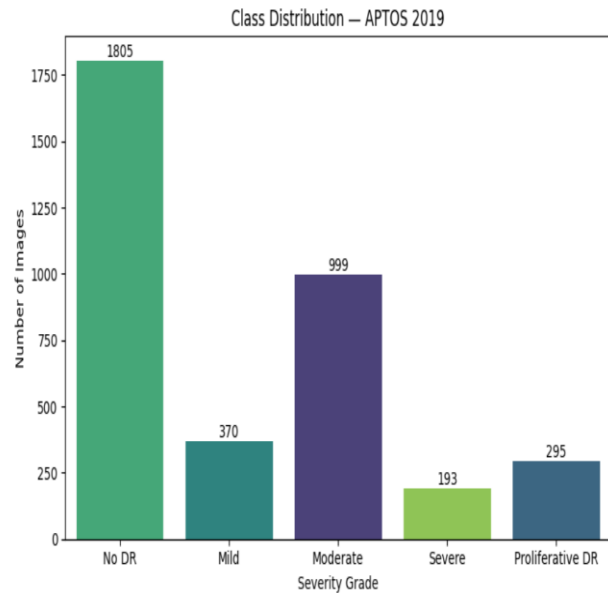


Fig: Class Distribution

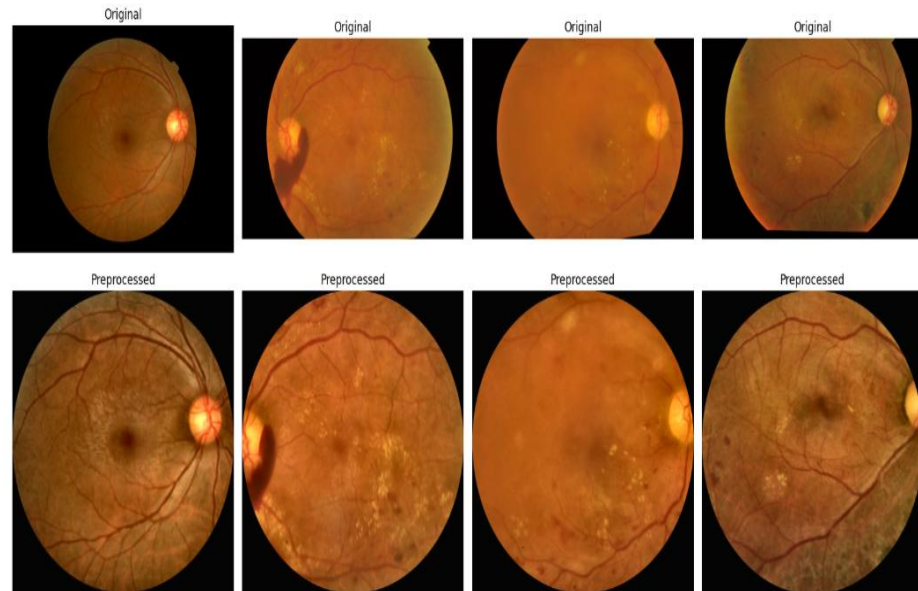


Fig : sample images from each class

References

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Any Queries?